

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO2

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 2

Unit Title: Search Techniques, Knowledge
Representation and Reasoning

Unit Mapped RBT: L2 (Understand)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain how Breadth First Search and Depth First Search differ, and state when each of the two should be preferred. (10 Marks)

- (a) Describe what each of the two expands first and the memory that each one requires.
 - (b) Explain which of the two you would choose when memory is tight and the solution may be deep, giving the reason.
- [L2 - Understand]

Q2. Describe the A* search algorithm and the part that each term of its evaluation function plays. (10 Marks)

- (a) State the evaluation function and describe what $g(n)$ and $h(n)$ each contribute.
 - (b) Explain what A* reduces to when $h(n)$ is zero everywhere, and what it reduces to when $g(n)$ is ignored.
- [L2 - Understand]

Q3. Explain what a heuristic function is and what makes one heuristic better than another. (10 Marks)

- (a) Describe what a heuristic estimates and state the condition it must satisfy for A* to remain optimal.
 - (b) Explain the exchange between the accuracy of a heuristic and the cost of computing it.
- [L2 - Understand]

Q4. Describe the Hill Climbing algorithm and the ways in which it can fail. (10 Marks)

- (a) Describe how it selects its next state and the condition on which it halts.
 - (b) Explain the local maximum, the plateau and the ridge, and state why the search stops short in each case.
- [L2 - Understand]

Q5. Distinguish uninformed search from informed search. (10 Marks)

- (a) Describe what each of the two knows about the problem, naming one algorithm of each kind.
 - (b) Explain what informed search gains by that extra knowledge, and what it must be supplied with in return.
- [L2 - Understand]

Q6. Describe a semantic network as a means of representing knowledge, and explain how inheritance works in one. (10 Marks)

- (a) Describe what the nodes and the links of a semantic network hold, using a small worked example.
 - (b) Explain how a property attached to a general concept becomes available to a specific one.
- [L2 - Understand]

Q7. Explain what a frame is and what it is able to hold about a concept. (10 Marks)

- (a) Describe the structure of a frame and what a slot may contain.
 - (b) Explain one kind of knowledge that a frame expresses more naturally than a semantic network does.
- [L2 - Understand]

Q8. Describe how reasoning under uncertainty is carried out using Bayes' theorem. (10 Marks)

- (a) State the theorem and describe what the prior, the likelihood and the posterior each represent.
 - (b) Explain, using a screening test for a rare condition, why a positive result need not make the condition likely.
- [L2 - Understand]

Q9. Explain the components of a production system and describe how one cycle of its operation proceeds. (10 Marks)

- (a) Describe the knowledge base, the working memory and the inference engine.

- (b) Explain what happens on each cycle, and why more than one rule may be ready to fire at the same moment.
[L2 - Understand]

Q10. Summarize the applications of the search algorithms studied in this unit. (10 Marks)

- (a) Describe two tasks that are naturally posed as a search problem, naming the states and the goal test in each.
(b) Explain which algorithm you would apply to each of the two, and give the reason.
[L2 - Understand]

----- End of Sample Question Set -----